TRINETRA-X

Finding planets in starlight — and being honest about it

A complete course: from "what is a star?" to defending it at ISRO

Team TRINETRA-X · BAH 2026 · Problem Statement 7

The big question

- Until 1995 we knew one planetary system. Now: ~6,000 exoplanets.
- Core questions: Are we alone? How common are Earth-like worlds? How do planets form?
- We are the first generation that can answer this with data.

The needle in a cosmic haystack

- A star is blinding; a planet is a tiny dark speck — we detect its EFFECT, not the planet.
- An Earth dims the Sun by only ~84 parts per million.
- Analogy: hearing one whisper in a stadium of 50,000 fans.

It's contrast, not telescope size — indirect methods sidestep the wall.

Four ways to find a planet

- Radial velocity — the star wobbles.   Transit — the star dims (← ours).
- Direct imaging — rare, big planets.   Microlensing/astrometry — gravity/position.
- Transit dominates: one camera monitors 10,000-100,000 stars at once.

The hidden enemy: computation

- You don't know the period → you must try thousands per star.
- Cost ~ (trial periods) x (data points) x (millions of stars).
- Most stars are EMPTY, yet blind search pays full price on every one.
- Analogy: a million locked doors, trying every key on each.

Light curves & transits

- Light curve = brightness vs time (a star's EKG).
- A planet crossing makes a small, repeating dip.
- Depth = (Rp/Rstar)^2 -> the planet-to-star size ratio.

Bug crossing a flashlight beam: dims a hair, then returns — on a schedule.

Dips have impostors — the U vs V tell

- A dip can be: planet / eclipsing binary / blend / junk (spots, systematics).
- Planet = rounded U (flat bottom). EB = pointy V, often + secondary dip + odd-even.
- THE KEYSTONE: depth alone can't tell them apart — SHAPE can.

Same symptom (a cough), different disease — you need more features to diagnose.

Depth, spacing, period

- Depth -> size. Spacing -> orbital period. Repetition -> reality.
- One dip could be a glitch; three evenly-spaced identical dips = a planet.
- Monotransits (one dip) are ambiguous -> out of Phase-I scope.

The incumbents: BLS & TLS

- BLS fits a box (fast, crude). TLS fits the real transit shape (sensitive, slow).
- TLS outputs SDE and is our gold-standard benchmark.
- Landmine: TLS's SDE is normalized over the grid you search (remember this!).

Birth of TRINETRA-X

- Blind TLS treats empty & promising stars identically.
- Idea: spend cheap effort to find EVIDENCE; spend expensive effort only where it exists.
- TRINETRA = 'third eye' — seeing what brute force misses.

Airport: metal-detector first; full search only on the people who beep.

The prime directive

- Find evidence first. Spend computation second. Let physics decide.
- Recall > precision: a false alarm is OK; a MISSED planet is not.
- Physics (depth/shape/repetition), not timing luck, is the arbiter.

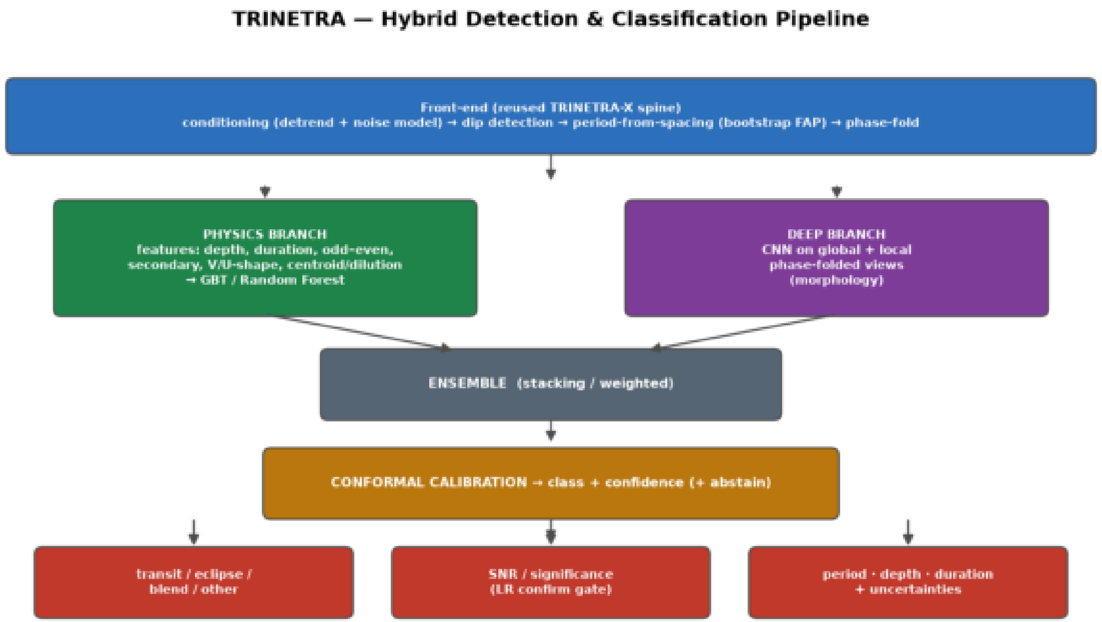
Cancer screen: never miss the tumor, tolerate false alarms.

The hypothesis (pre-registered)

- H1: routing cuts compute vs TLS WITHOUT losing recall. H0: it doesn't.
- E1 = recall non-inferiority (lose no planets). E2 = compute reduction >= 30%.
- Frozen BEFORE touching test data — that's what makes the answer trustworthy.

Architecture

- Condition -> detect events -> period from spacing -> bootstrap FAP -> Lambda confirmer -> TLS fallback.
- Savings come from folding k EVENTS, not searching N points x a period grid.



Module deep-dive

- Detector: cheap box matched filter, SNR = depth / scatter.
- Period-from-spacing: read P off event spacing (Rayleigh Z = k R^2).
- Bootstrap FAP: is the periodicity real or lucky noise? (block bootstrap).
- Confirmer: likelihood ratio Lambda = the PHYSICS arbiter. Fallback: if unsure, full TLS.

Five math anchors

- 1) delta = (Rp/Rstar)^2 — depth <-> size
- 2) SNR1 = delta / CDPP — detectability
- 3) FAP = (1/B) sum 1[T_b >= T_obs] — honesty/calibration
- 4) Lambda = -2 ln[L(flat)/L(transit)] — the judge
- 5) saving ~ f , recall loss = -f(1 - r_seed g) — the whole bet

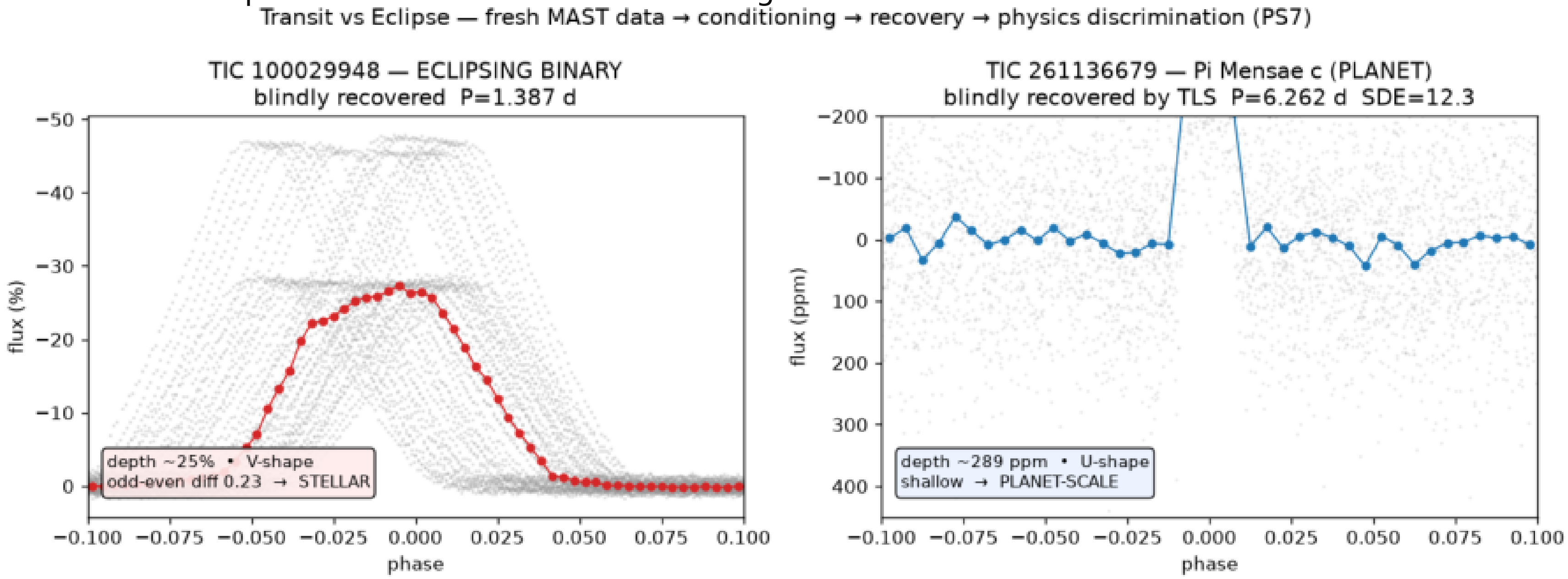
The research journey

- Idea -> build & calibrate (M0-M3, sealed) -> CRISIS: Finding B.
- Finding B: TLS SDE is grid-normalized -> 'targeted TLS' design is invalid.
- Repair: v3 -> Lambda confirmer at a COMMON false-alarm rate (re-registered openly).
- Tried to cheapen the bootstrap; an equivalence test said no -> kept it.

We caught a flaw in our own case BEFORE the verdict and fixed it in the open.

The verdict (read once)

- TEST read exactly once; verdict pre-committed.
- E1 PASS: recall preserved (delta-recall -0.48 pp).
- E2 FAIL: compute reduction 24.4% < 30%.
- => H1 falsified on the compute branch — a SUCCESSFUL negative result.



Why it failed + the scaling insight

- Killer: un-cheapenable B=1000 bootstrap 'entry tax' (rho_d~14.4%) + fallbacks.
- Theory: break-even prevalence pi* = rho_d / f_p -> short baselines can't win.
- Insight: fast lane folds k events while TLS cost grows with baseline.
- => the advantage should GROW with search-space size -> test on Kepler.

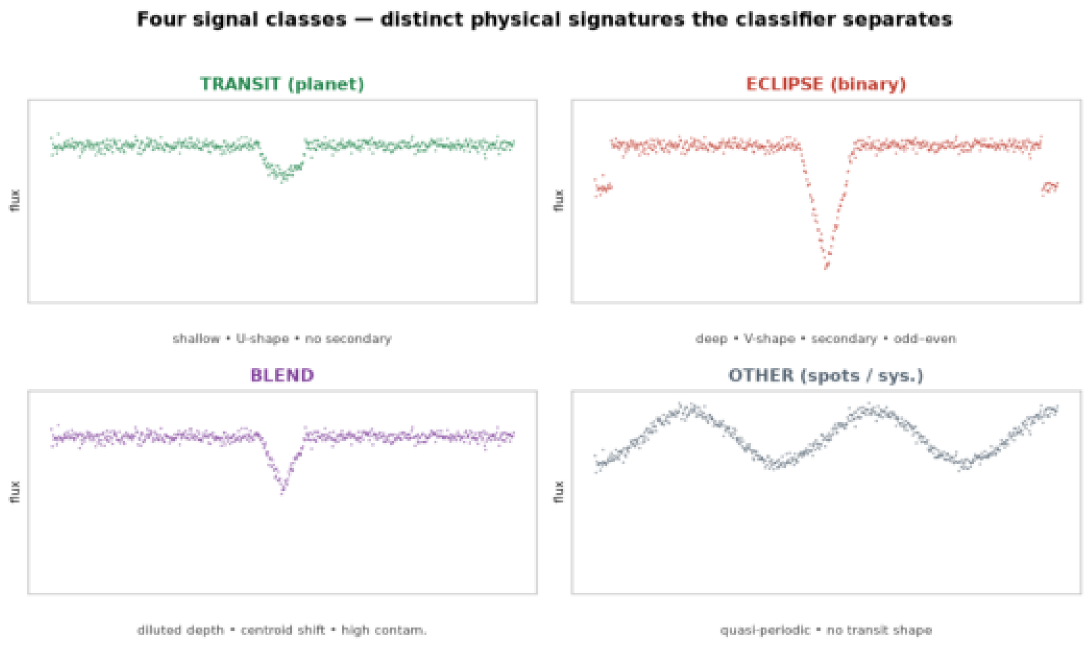
A shortcut that only pays off on a long highway, not a short street.

Milestones M0-M7

- M0 sealed dataset + leak-safe split. M1 conditioning. M2 chose 2.5-d window by injection.
- M3 calibrated thresholds (+ cleaned EB-contaminated null pool).
- M4 the single sealed test (the verdict).
- M5-M7 characterization + reality check (real TOIs: Arm B = Arm A = 86.7%) + manuscript.

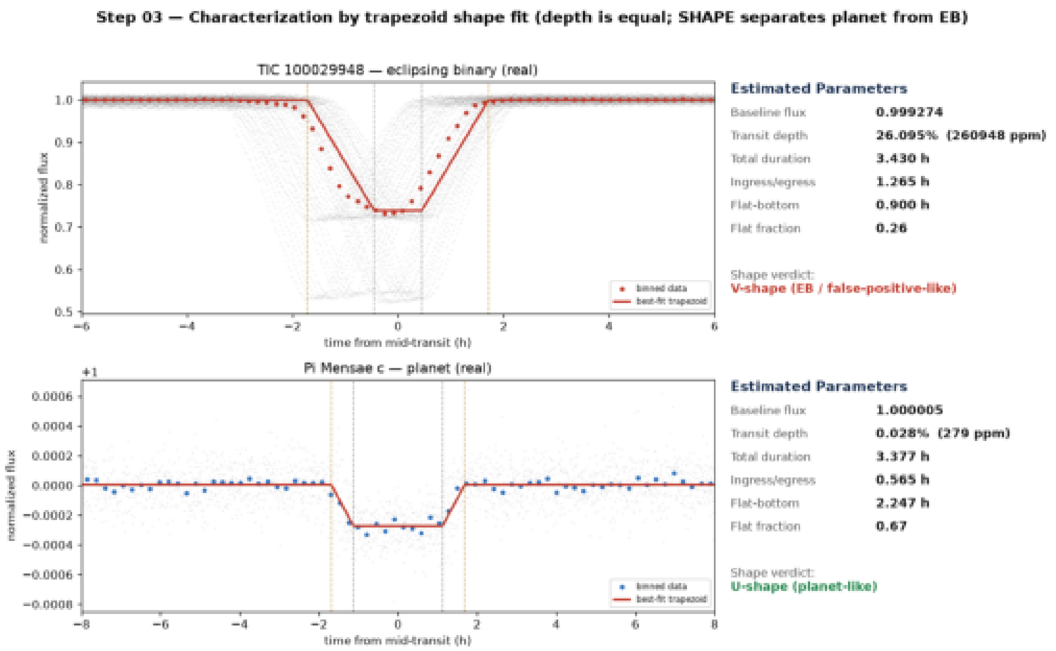
The hackathon pivot (PS7)

- Same validated spine, new mission: CLASSIFY dips for ISRO PS7.
- 5 steps: detrend -> identify -> characterize -> classify -> significance.
- Committee's key point = our keystone: depth doesn't separate planet from FP; shape does.



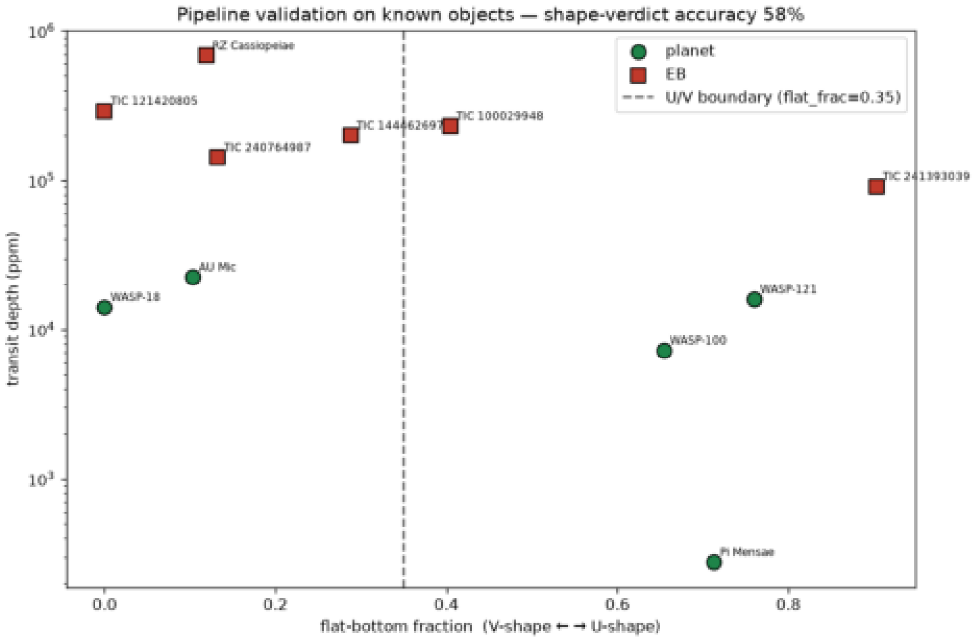
Characterization: U vs V (real data)

- Trapezoid fit reproduces the committee's slide-5/6 parameters.
- EB (TIC 100029948): flat-fraction 0.26 -> V (false positive).
- Pi Mensae c: flat-fraction 0.67 -> U (planet). Depth nearly equal; SHAPE decides.



Classification + validation

- Physics features -> gradient-boosted classifier, conformal-calibrated confidence.
- Validated on 12 known objects: planets U + literature periods; EBs V.
- Single feature = 0.58, but classes separate in 2-D depth x shape -> multi-feature justified.



Current status + roadmap

- Phase I sealed/final (no v4). Round-1 package complete (submit by 2026-07-01).
- Round 2: robust period recovery, pixel-level centroid for blends, CNN ensemble, full report.
- Research future: Kepler scaling experiment.
- Named risk: period recovery on active/short-period stars.

The story in one slide

- Humanity learns to hear planets by watching stars blink.
- The gold-standard detective interrogates everyone; a 'third eye' glances first.
- A hidden statistical trap nearly fakes a failure; we catch & fix it openly.
- One honest look: 'planets kept, savings missed' -> a bigger stage (Kepler).

Failure with a lesson is how science moves forward.

Pitch ladder

- Practice 7 lengths: 30s / 1m / 3m / 5m / 10m / 15m / 30m.
- Default for judges: the 2-minute judge script.
- Always open with the honesty spine; always close on the forward path.

Master summary — why it matters

- One hypothesis, one architecture, one honest verdict, one forward path.
- Rare: a project with the integrity to falsify its own headline while proving its core works.
- It produced BOTH a future experiment (Kepler scaling) AND a working tool (PS7 pipeline).

Intuition + mathematics + honesty — that's what makes you the expert on your own work.